

A Technical Introduction to CO₂ Emissions & Monitoring

A decorative graphic on the right side of the slide. It features a light blue line that starts at the top left, goes down to the middle right, and then curves down to the bottom right. Along this line are five colored dots: a teal dot at the top left, an orange dot at the middle right, a red dot at the bottom right, a blue dot at the middle left, and a green dot at the bottom right. Each dot is enclosed in a thin white circle.

From global inventories to facility-level decisions

38.1

Gt CO₂ in 2023
fossil + industry

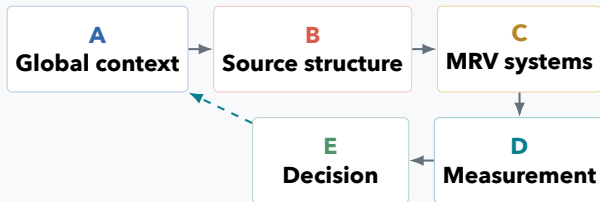
68%

increase since 1990
territorial total

3 layers

inventories, observations,
inverse modelling

The analytical chain: from signal to action



feedback: targets, controls, redesign

Core question: What evidence is sufficiently accurate, timely and attributable for the decision being made?

THREE EVIDENCE LAYERS

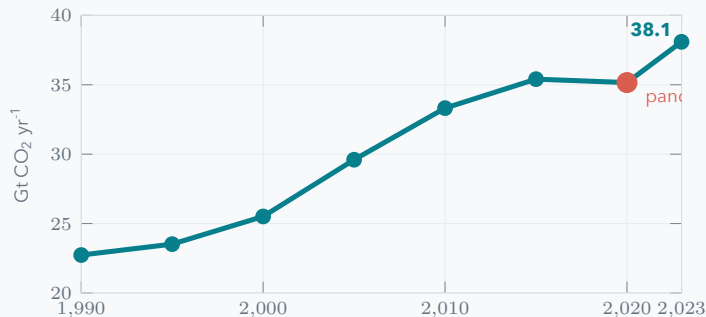
- 1. **Estimate** activity $\times$ factor
- 2. **Observe** concentration / proxy
- 3. **Reconcile** model + uncertainty

Learning outcomes

- Interpret emissions datasets
- Design an auditable MRV chain
- Match methods to decisions

Source: IPCC TFI; GHG Protocol Corporate Standard.

A. Global fossil CO₂: growth interrupted, not reversed



+68%

1990-2023

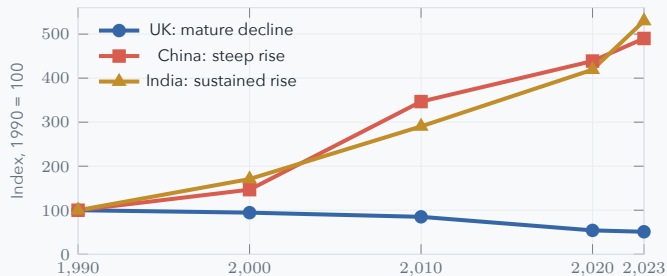
+5.36

Gt since 2010

A temporary shock changed the *level*; it did not establish a sustained downward *trend*.

Source: Global Carbon Budget (2025), processed by Our World in Data. Territorial fossil and industrial CO₂; excludes land-use change.

A. Three national pathways, one global accounting problem



**DO NOT INFER CAUSALITY FROM
SHAPE ALONE**

- Industrial structure
- Energy access and demand
- Trade displacement
- Policy and technology

Normalize for trajectory; retain absolute tonnes for climate relevance.

Source: Global Carbon Budget (2025) via OWID. Values are territorial fossil and industrial CO₂.

A. The metric defines the claim

BOUNDARY

Territorial

Inside national borders

TIME BASIS

Annual

◀ Current flow and policy

Consumption

Adjusts for embodied trade

Cumulative

◀ Historical contribution

System scope: fossil + industry vs. land-use change

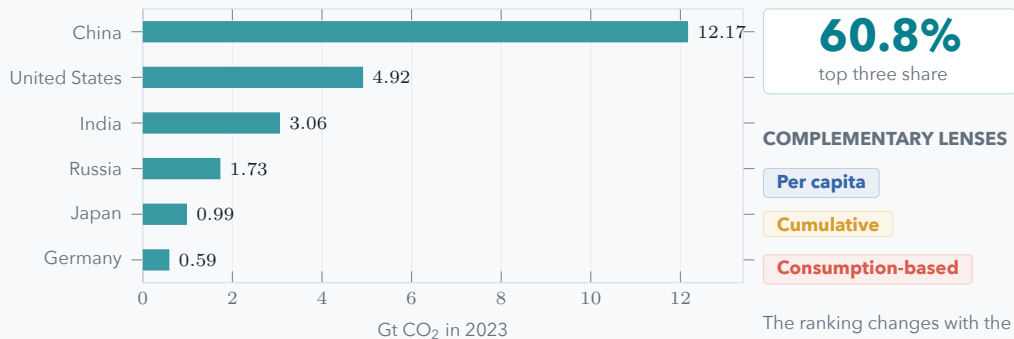
A DEFENSIBLE CHART STATES

1. Gas and warming metric
2. Organizational/geographic boundary
3. Time period and revision
4. Included/excluded sources

"CO₂ emissions" is not a complete variable definition.

Source: OWID data sources and methods; Global Carbon Budget FAQ.

B. Major emitters: scale is concentrated, responsibility is multidimensional

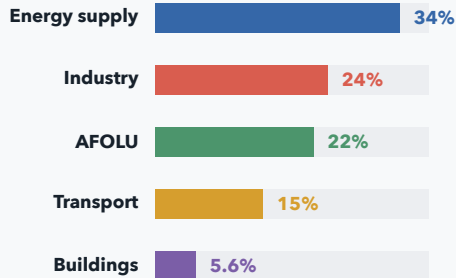


The ranking changes with the policy question; the atmosphere responds to the total.

Source: Global Carbon Budget (2025) via OWID; world total = 38.094 Gt CO₂.

B. Sector structure: direct emissions and electricity attribution

DIRECT GLOBAL GHG EMISSIONS, 2019



Direct allocation

Power generation remains in energy supply.

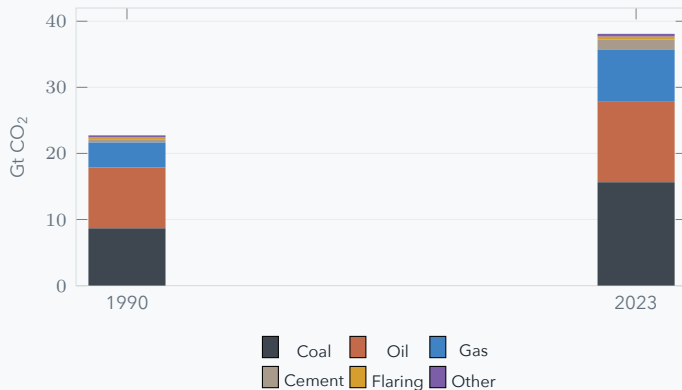
End-use allocation

Electricity and heat move to consuming sectors.

Result: industry rises from 24% to 34%; buildings from 6% to 16%.

Source: IPCC AR6 WGIII, Ch. 2; sector shares are GHG in CO₂e, not CO₂ alone.

B. Fuel decomposition: coal remains the largest component



41.1%

coal share, 2023

32.2%

oil share, 2023

20.5%

gas share, 2023

Fuel totals expose the physical levers behind the aggregate trend.

Source: Global Carbon Budget (2025) via OWID fuel series. Values may not sum exactly due to rounding.

C. MRV is an evidence architecture, not a single dataset



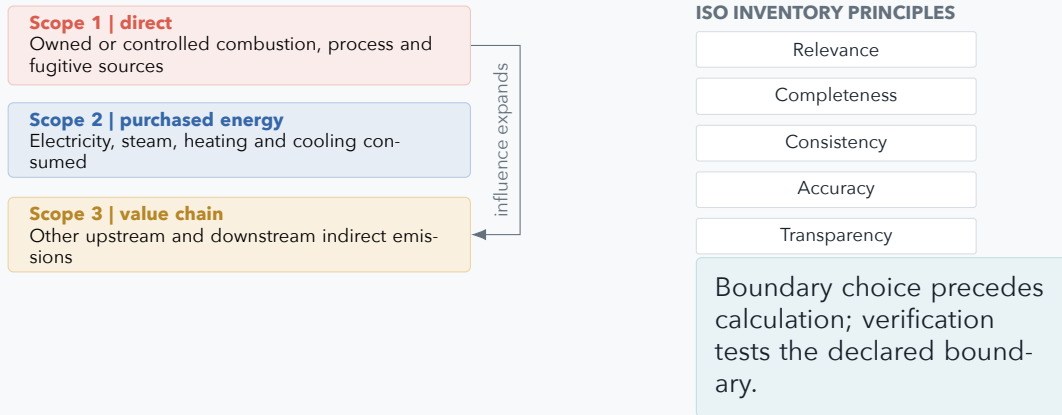
MINIMUM AUDIT FIELDS

- Source identity and owner
- Method, factor and version
- Unit and temporal coverage
- Missing-data treatment
- Reviewer and evidence link

A precise number without provenance is weak evidence.

Source: GHG Protocol; ISO 14064-1:2018; IPCC TFI.

C. Corporate boundaries: GHG Protocol and ISO 14064-1



Source: GHG Protocol scope definitions; ISO 14064-1:2018.

C. Verified registries convert calculations into compliance evidence

EU ETS / Union Registry

Approved monitoring plan → accredited verification → verified emissions and allowance surrender

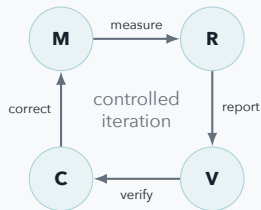
US EPA GHGRP

Facility/source-category methods; activity, mass balance or CEMS; public facility-level reports

National inventories

IPCC methods, category completeness, time-series consistency and review

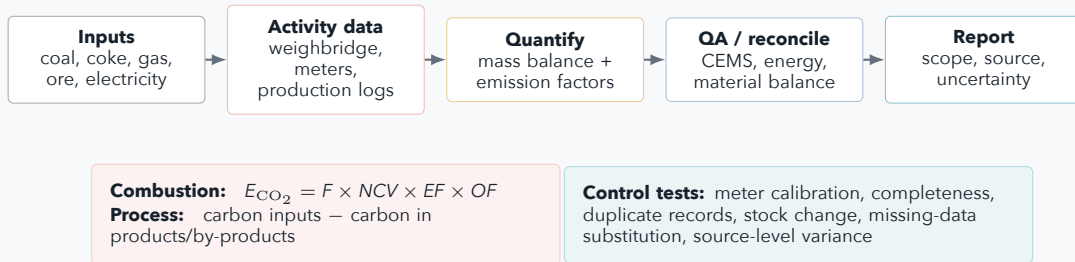
ANNUAL COMPLIANCE CYCLE



Verification is risk-based assurance, not proof of zero error.

Source: EU ETS MRV; US EPA GHGRP; IPCC TFI.

C. Example: auditable CO₂ workflow for a steel facility



Source: US EPA GHGRP Subpart Q; Climate TRACE heavy-industry methods; GHG Protocol Corporate Standard.

D. Bottom-up techniques: choose the closest measurable driver

Method	Best use	Dominant limitation
Fuel meter	Combustion sources	Calibration, fuel quality
Mass balance	Carbon-rich processes	Stock and carbon content
CEMS	Large stacks	Representativeness, downtime
Production proxy	Rapid estimates	Technology heterogeneity
Emission factor	Broad inventories	Transferability, vintage

Higher-tier methods replace generic factors with source-specific activity, composition or direct measurement.

GENERIC MODEL

$$E = \sum_i A_i EF_i \eta_i$$

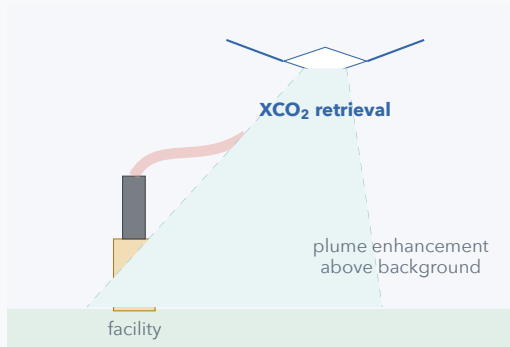
activity × factor × control

Design principle

Use the simplest method that meets the required materiality, frequency and attribution.

Source: IPCC inventory methodologies; IPCC Emission Factor Database; US EPA GHGRP.

D. Satellites observe atmospheric columns, not ledger entries



WHAT THE SIGNAL REQUIRES

Column concentration

Wind / transport

Background estimate

Cloud / aerosol screening

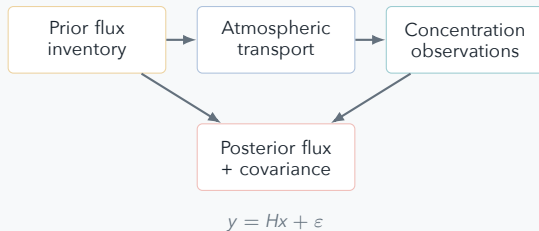
OCO-2/OCO-3: CO₂ column retrievals.

Sentinel-5P: daily trace-gas context (CO, NO₂, CH₄, etc.), not a CO₂ product.

Source: NASA OCO-2; Copernicus Sentinel-5P; Nassar et al. (2022), doi:10.3389/frsen.2022.1028240.

D. Top-down inversion and near-real-time nowcasting answer different questions

ATMOSPHERIC INVERSION



CARBON MONITOR NOWCAST



Power generation, industry, transport, buildings and aviation scaled with timely activity proxies

Nowcasts optimize latency; inversions optimize atmospheric consistency.

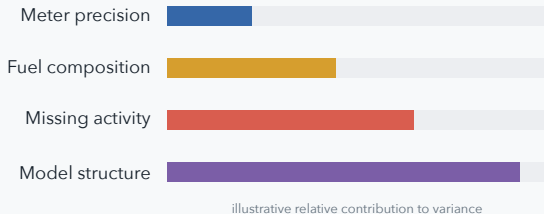
Source: NOAA CarbonTracker; Liu et al. (2020), Carbon Monitor; carbonmonitor.org.

D. Uncertainty is structured information

PROPAGATION FOR A SIMPLE PRODUCT

$$\left(\frac{\sigma_E}{E}\right)^2 \approx \left(\frac{\sigma_A}{A}\right)^2 + \left(\frac{\sigma_{EF}}{EF}\right)^2$$

valid for independent, approximately small errors



REPORT MORE THAN $\pm x\%$

- Random error vs. systematic bias
- Correlation across sources and time
- Coverage and detection limits
- Version and revision uncertainty
- Sensitivity to model assumptions

Triangulation detects disagreement; it does not automatically identify which estimate is correct.

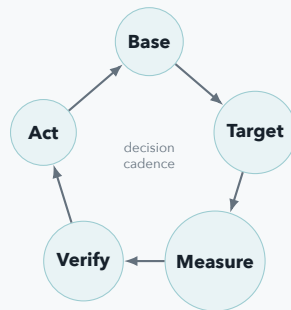
Source: IPCC TFI uncertainty FAQ; Global Carbon Budget 2024 reports fossil CO₂ uncertainty near $\pm 5\%$ and substantially larger land-use uncertainty.

E. Match the monitoring system to the decision

Decision	Primary evidence	evi-	Critical quality
National target	Inventory + review		Completeness, consistency
Carbon market	Verified MRV	facility	Accuracy, legal traceability
Plant operation	Meter/CEMS now-cast		Latency, source attribution
Supply chain	Supplier data	primary	Boundary and allocation
Anomaly detection	Satellite + model		Coverage, false positives

The “best” system is the least complex evidence chain that changes the decision reliably.

CLOSED-LOOP ACTION



Source: Synthesis of IPCC TFI, GHG Protocol, EU ETS MRV, NASA/NOAA and Carbon Monitor methods.

E. Takeaway: build a reconciled, decision-grade evidence chain

1. Define the variable. Gas, boundary, time basis and exclusions.

2. Preserve provenance. Every result links to activity, factors and versions.

3. Triangulate. Bottom-up accounting, observations and models test different failure modes.

4. Design for action. Accuracy, latency and attribution are decision-specific.

REFERENCE MAP

Global totals

Global Carbon Budget
Our World in Data

Sector and methods

IPCC AR6 WGIII
EDGAR / JRC

Corporate and compliance

GHG Protocol
ISO 14064-1
US EPA GHGRP
EU ETS

Observations and rapid estimates

NASA OCO-2 ; NOAA CarbonTracker
Carbon Monitor ; Climate TRACE

**Good monitoring does not merely count carbon.
It makes claims testable and interventions accountable.**